MANAN TUTEJA

MECHANICAL/AEROSPACE ENGINEER

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SUMMARY

Mechanical and Aerospace Engineering student interested in design and controls. Developed technical and interpersonal skills through active participation in clubs and coursework. Committed to continuous learning and professional growth in engineering.

TECHNICAL SKILLS

- CAD design: Onshape, SolidWorks, Autodesk Fusion 360, Autodesk Inventor
- Manufacturing processes: Lasercutting, Waterjetting, 3D printing
- Arduino
- Project management with Notion
- Team leadership skills
- Robotics engineering expertise
- MATLAB and Simulink: Control system modeling
- Python programming: OpenCV, ROS2
- Hobbyist PCB Design: EasyEDA

EXPERIENCE

CENG/NANO 15R Reader, UCSD CENG **Sep 2025 - Dec 2025**

- Reader/Grader for Engineering Computation Using MATLAB

MAE 8 Reader, UCSD MAE **Sep 2024 - Dec 2024**

- Reader/Grader for MATLAB Programming for Engineering Analysis class.

Mechanical Team Lead, Club: Triton Robotics **Jan 2024 - Present**

- Led mechanical subteam in robot design and manufacturing initiatives.
- Cascaded Position and Velocity PID controller tuning and modeling using Matlab/Simulink
- Developed comprehensive CAD and manufacturing training sessions covering Onshape, 3D printing, laser cutting, and waterjetting.
- Engineered yaw mechanism employing belt and pulley system using Onshape.
- Designed tangential suspension for mecanum wheel X-drive chassis configuration utilizing Onshape.
- Created and integrated octagonal X-drive omniwheel chassis using Solidworks.
- Designed linear suspension system emphasizing cost-effectiveness and assembly efficiency with Solidworks.

Design Engineer; Club: First Robotics Competition (FRC) **Aug 2020 - Jun 2023**

- Design, deployment, and manufacturing of 3D printed differential bevel geared wrist using Fusion 360. Designed tank drive base and led turret manufacturing.

Chief Scientist, Class/Club: Exagora Engineering **Aug 2022 - Jun 2023**

- Led design, production, and deployment of 12-module-long 3D-printed robotic modular snake using Feetech servos. (Fusion 360)
- Design of 12KG lift drone

EDUCATION

M.S. Mechanical Engineering **Sep 2026 - Jun 2027**
University of California San Diego

B.S. Aerospace Engineering **Sep 2023 - Jun 2026**
University of California San Diego

- 3.822 Cumulative GPA
- 3.728 Major GPA

Relevant Coursework:

AWARDS/ACHIEVEMENTS

- Beach Blitz Regional Winner - FRC (2023)
- Air and Space Force Association StellarXplorers Competition VII Finalist (2022)
- National Merit Scholarship Finalist (2023)